Lucas Leong

+65 8189 9220 | leonghongkit@gmail.com | linkedin.com/in/lucas–leong–hk | hrtowii.dev

Experience

Software Engineer Intern Aug. 2025 – Oct. 2025

Precix *precix.io*

- Explored reverse engineering of proprietary ultrasound firmware DLLs and OpenCL kernels to study real-time medical imaging pipelines using IDA Pro and windbg
- Developed embedded C firmware for IMUs and ultrasound probes, implementing multithreaded data acquisition and wireless transmission on Raspberry Pi CM5 and ESP32-P4/C6.
- Prototyped end-to-end system integrating sensor and imaging data into an ElysiaJS backend with a Svelte-based frontend for real-time visualization using websockets.

Software Developer Feb. 2024 – May. 2024

eas-ai.com

- Developed a full-stack web application using NextJS, Prisma, PostgreSQL and Docker, providing a platform for funeral parlors to sell funeral packages for customers

Projects

Serotonin | *Objective-C, C, iOS Kernel Exploitation* Jan. 2024

- Independently researched and implemented a novel iOS 16.0–16.6.1 semi-jailbreak; published open source, gaining **650+ GitHub stars** and **65,000+ downloads**
- Developed original kernel namecache vnode redirection technique to hijack launchd execution without breaking SSV, bypassing iOS kernel memory protections
- Reverse engineered iOS kernel internals (vnodes, PAC, launch constraints, codesigning) using IDA Pro and kernel read/write primitives from kfd exploit chain

Wayside – Hack&Roll 2026 Winner | *C, ESP32, ESP-NOW, BLE, Embedded Systems* Jan. 2026

- **Won top 9**; built proximity-based networking hardware badges inspired by Nintendo StreetPass for passive discovery of compatible hackathon participants
- Implemented custom network stack from scratch using ESP-NOW for peer broadcasting and RSSI-based proximity estimation on ESP32 microcontrollers
- Integrated BLE pairing with personal devices and cryptographic layer for secure peer-to-peer data exchange between badges

Balatro Neural Network | *Python, Lua, Deep Q-Learning, Smods* Feb.–May 2025

- Built a game mod using Smods and Lovely Injector exposing internal game state to a Python server over a custom client-server protocol
- Trained a Deep Q-Learning agent to make real-time in-game decisions; agent learned basic card selection strategies after training

BrainHack – TIL-AI | *Python, Docker, TensorFlow* June 2023

- Achieved **8th place** in semifinals; finetuned vision models for zero-shot object detection deployed via Docker Compose onto physical robot hardware

Skills

Languages: C, Python, JavaScript, TypeScript, Bash, Lua, Kotlin, Rust
Systems & Security: IDA Pro, pwntools, gdb, WinDbg, ESP-IDF, Linux, Docker
Web: React, NextJS, Node.js, Bun, Elysia, Flask, FastAPI
Tools: Git, Vercel, Figma

Education

Nanyang Polytechnic Singapore, Singapore

Diploma in Information Technology *Apr. 2023 – Apr. 2026*

Singapore University of Technology and Design (SUTD) Singapore, Singapore

Computer Science and Design *2026 – 2029*